

University of West
London
School of Computing &
Engineering



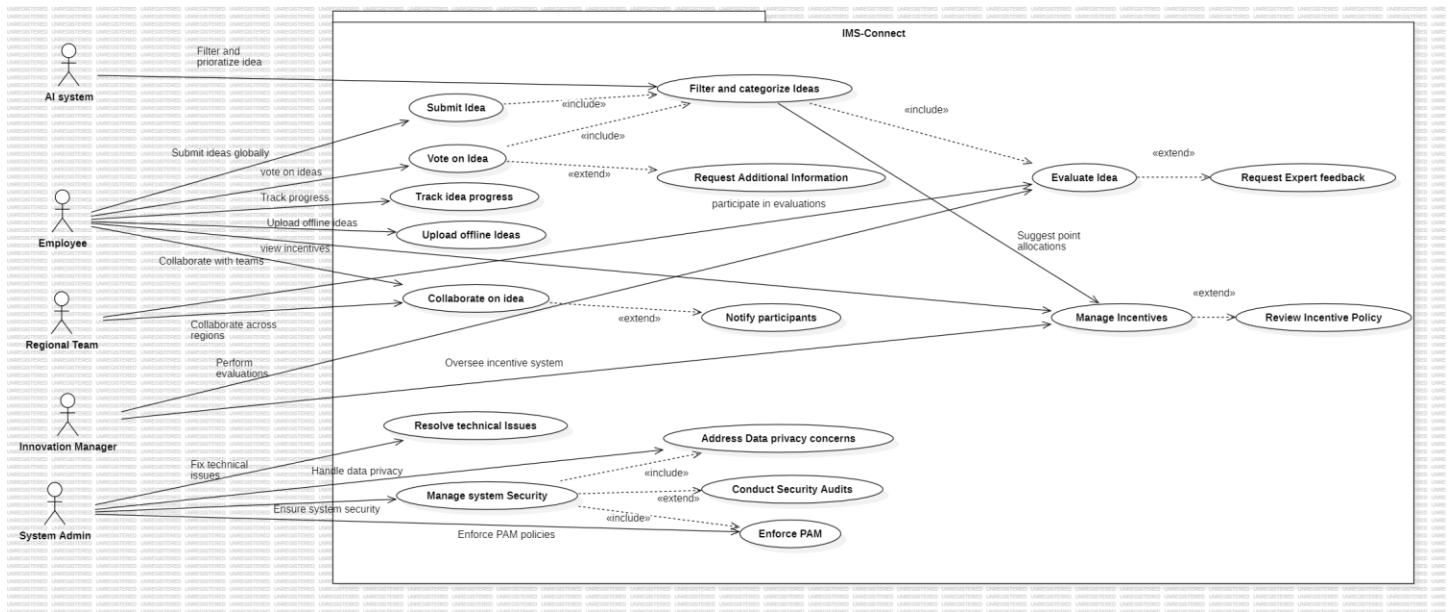
-Component Model-

Module	Mobile Web Component Development
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Module Leader:	
Student Name:	
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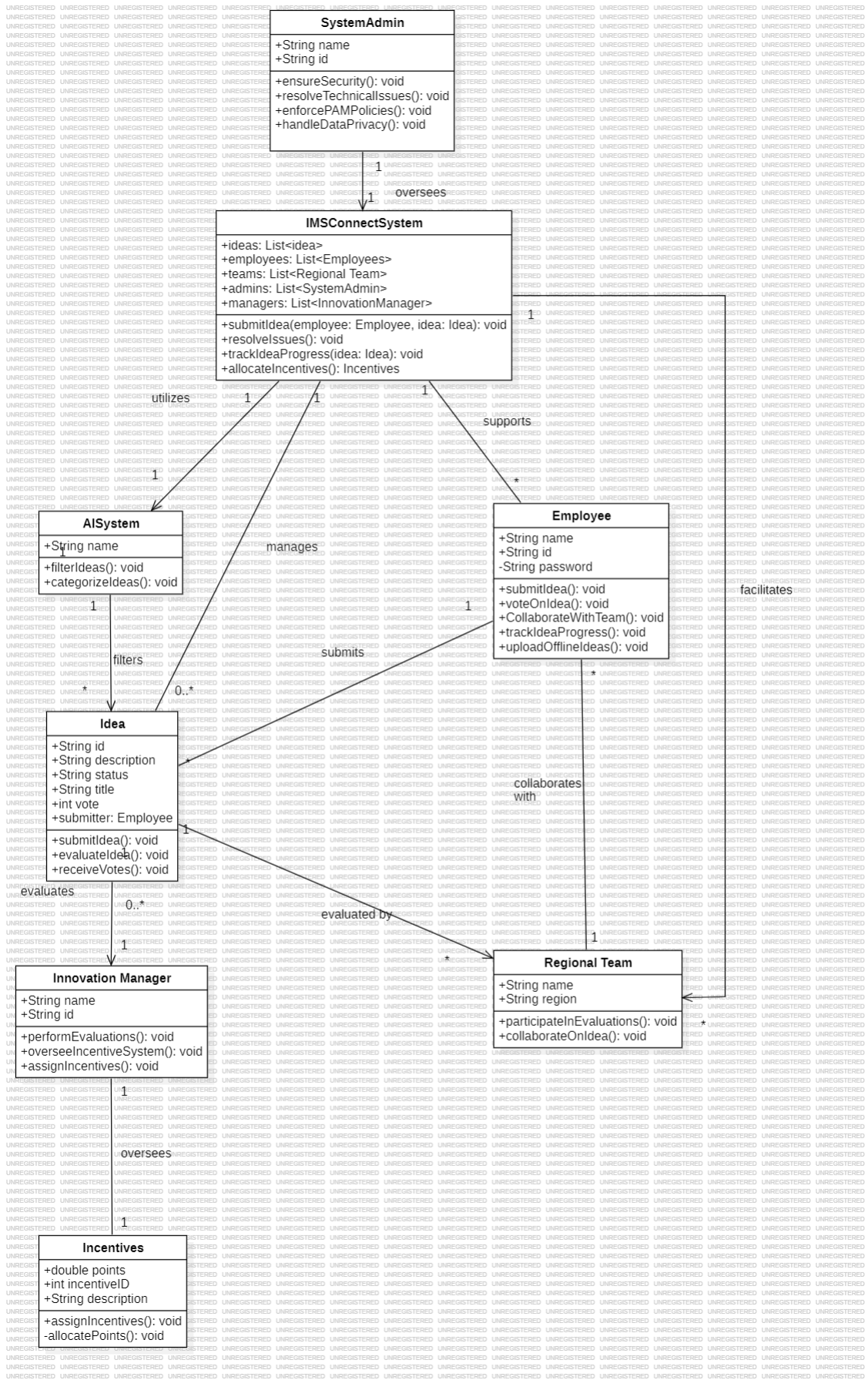
1. Use Case Model



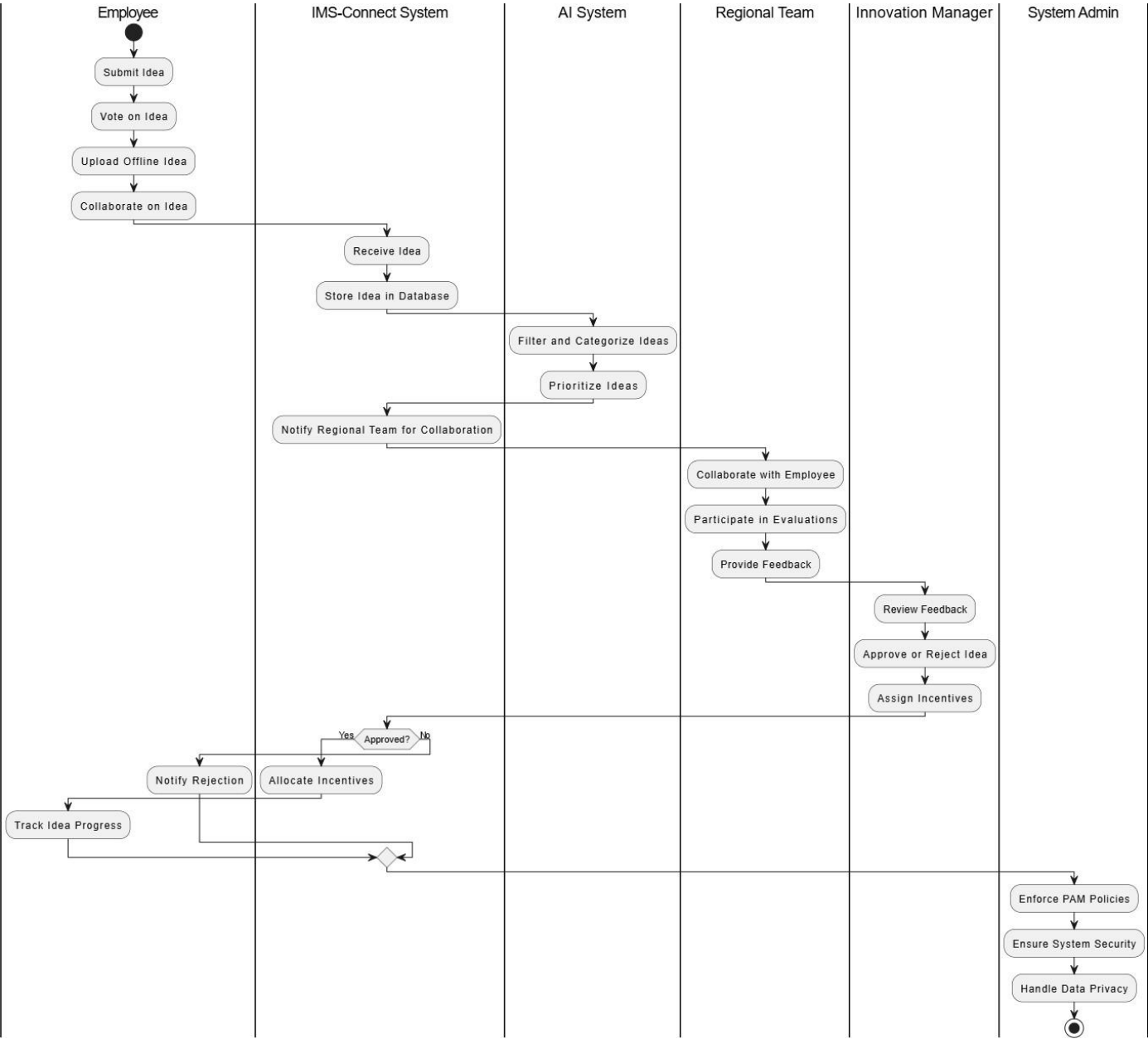
The above Use case diagram for IMS-Connect system highlights the interactions and functions between the system's actors, namely, Employee, Innovation Manager, System Admin, Regional team and AI System. The employee is the primary contributor in the idea generation and voting process. They are involved in submitting ideas, voting on them, collaborate with teams to work together and develop innovative solutions and track their progress, also they can submit offline ideas when connectivity issues arise and also can view the incentive system. The Regional team mainly collaborate with employees across the regions and participates in evaluation process to provide insights on submitted ideas. The Innovation Manager oversees evaluations and manages incentive system. This may extend to reviewing the submitted ideas, review incentive policy etc. which helps to ensure that the overall process aligns with organizational goals. Ensuring system security, resolving technical issues and handling data privacy and enforcing Privileged Access Management (PAM) policies is done by System Admin. Managing system Security may include address data privacy concerns and enforcing PAM as well while it could also extend to performing security audits. Finally, the AI system, filters and categorizes the submitted ideas using AI powered solutions which will help to prioritize ideas for further evaluation.

Therefore, by above, use case descriptions and explanations of interactions are described.

2. Business Concepts

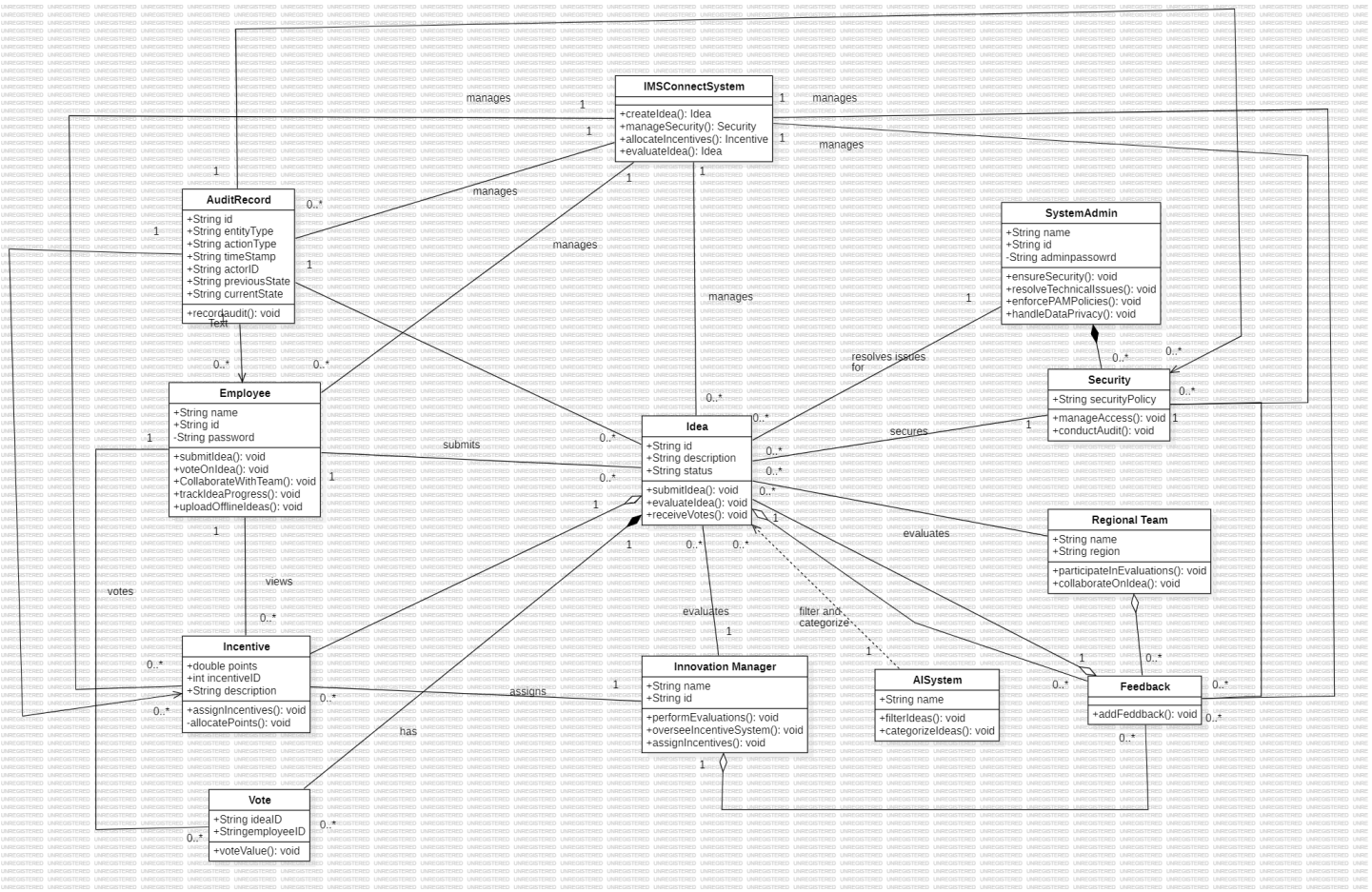


3. Business Process Model



4. Type Model

Below shows a comprehensive type model class diagram with core types and business component. This also includes an audit table showing refactoring from the Business Concept model.



5. Component Specification

5.1 Business Rules

Rules for Idea Submission:

- Duplicate idea submission is not allowed
- Each idea should be automatically submitted with a unique ID along with a timestamp of the submission.
- For a successful submission of an idea, title, description and region needs to be entered.
- Offline submission is allowed for regions with connectivity issues.

Rules for Collaboration:

- The projects flagged as High priority need to have a cross regional collaboration mandatorily.

Rules related Voting:

- Votes must be updated in real time
- An employee cannot modify their votes after submission and can vote only once
Self-voting is prohibited by restricting voting to ideas that are not submitted by the same employee.

Rules for Incentive:

- Points are awarded for the submission of ideas, collaboration, and voting in accordance with predetermined thresholds:
 - 10 points for submitting an idea.
 - Working together on a concept that has been approved: 20 points.
 - Voting for ideas is worth two points, with a monthly maximum of ten points.
- Only Innovation managers can assign incentives.

Rules for Tracking:

- Each concept needs to go through the following phases: Submitted -> reviewed -> accepted -> denied -> in progress/ archived

Rules for AI System:

- Within a time period of 24 hours, AI should categorize and filter the submitted ideas.
- The ideas with an AI score below 25% will be archived after the review by managers.

Rules for Data Security:

- The AuditLog should contain the details such as actor, actions, and timestamps etc.
- Innovation Managers and system admin can only modify the status of ideas.
- Restrictions must be applied to all sensitive data and processes by the PAM.

5.2 System Interfaces

Employee Interfaces:

- Allows employees to submit ideas, vote, track progress, and view rewards
- Language can be selected based on the region.
- Web portal and mobile can be used to access the system

Regional Team Interface:

- Allows access for team collaboration, and idea evaluation

Innovation Manager Interface:

- Allows to review ideas, assign incentives and manage the overall progress.
- For high priority ideas approval workflows are included

Voting Interface:

- Employee votes are tracked and database is updated in real time
- Integrated with employees and regional Interfaces.

PAM Security Interface:

- For audit purposes, all privileged actions are logged
- Checks for authentication and authorization for access of the system

Interface for Offline Submission:

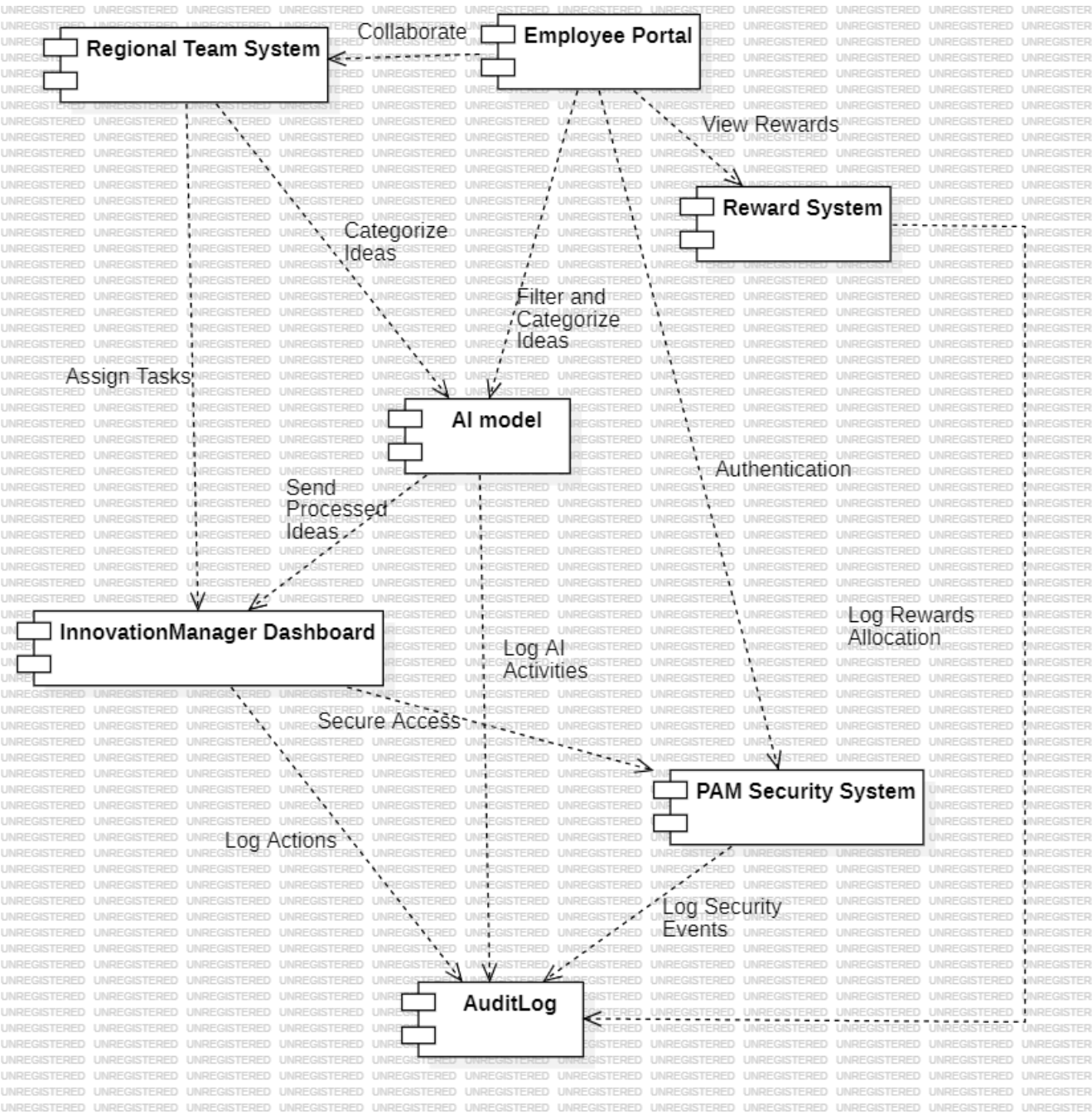
- Collects ideas during low connectivity times and updates database when connectivity is restored.

AI system Interface

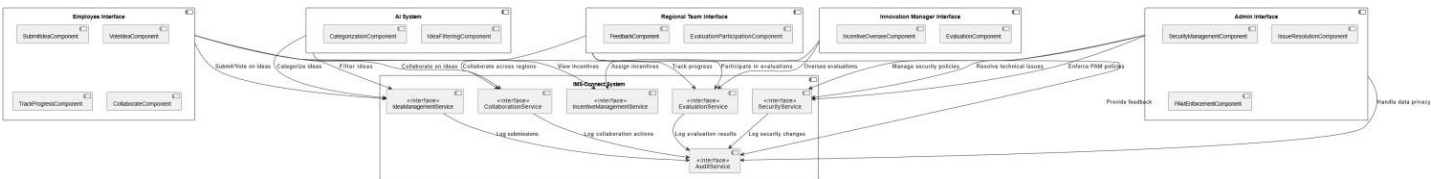
- This is used for idea filtering and categorization which is automated.
- Gives results on priority details, and patterns.

6. Component Architecture

6.1 Component Architecture

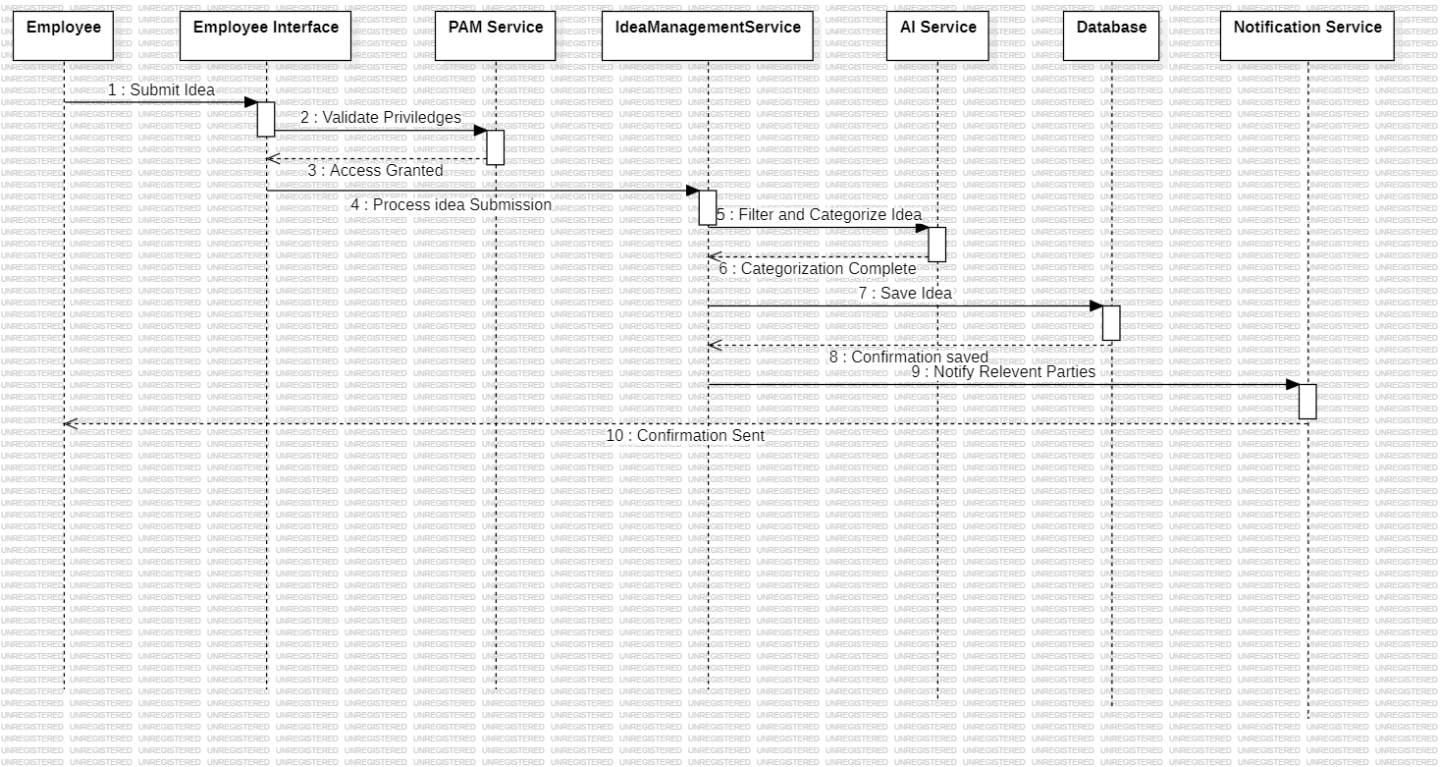


Below is a detailed component diagram drawn of the system



6.2 Interaction Diagrams

Sequence Diagrams are drawn as below to show interactions.



7. Assumptions

The following assumptions are made in-order to construct the model.

- The system will be web-based with offline submission options for low-connectivity regions
- Employees will authenticate using company credentials, and role-based access will be enforced.
- The idea evaluation process will be overseen by Regional innovation managers.
- The company uses English as the default language, with optional support for regional languages.
- Ideas will be prioritized based on votes, innovation manager input, and system flags (e.g., AI-based categorization).
- System administrators are in charge of maintaining system security, resolving technical problems, implementing PAM regulations, and managing privacy issues pertaining to data.
- A single employee can only vote once on each idea.